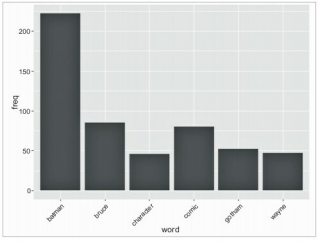

Figure 2: Depicts the plot of words with a frequency greater than 30

```
  batman   bruce   comic   gotham   wayne character
     222      85      80       52      47        46
```

Finding associations

After the previous step, we get a rough idea that the terms 'batman', 'bruce', 'comic', 'wayne' and 'character' are central to our corpus. Thus, if we want to find terms that highly correlate with them we can use the *findAssocs()* function, which takes as its parameter the DTM, the word and the correlation limit (which ranges from 0 to 1). A correlation of 1 means 'always together', while a correlation of 0.5 means 'together for 50 per cent of the time'.

```
findAssocs(dtm, "bruce", corlimit=0.90)
$bruce
   able    pain   times   order    will    time
   0.99    0.98    0.98    0.97    0.96    0.95
 access     day   defeat   earth    eyes    idea
   0.94    0.94    0.94    0.94    0.94    0.94
individual  minds  nightwing  sometimes  unlike  character
   0.94    0.94    0.94    0.94    0.94    0.93
associated    life    part personality   alfred     man
   0.92    0.92    0.92    0.92    0.91    0.91
parents
   0.91
```

Plotting the term frequencies

To graphically represent our term frequencies as a plot, the ggplot2 library is used as follows:

```
library(ggplot2)
wf <- data.frame(word=names(freq), freq=freq)
p <- ggplot(subset(wf, freq>30), aes(word, freq))
p <- p + geom_bar(stat="identity")
p <- p + theme(axis.text.x=element_text(angle=45, hjust=1))
p
```

Figure 3: Word Cloud of words with a frequency greater than 30

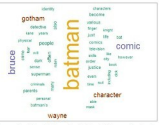

Figure 4: Word Cloud of the 50 words that occur most often in the document

Word Cloud

Word Cloud is another way of representing the frequency of terms in a document. Here, the size of a word indicates its frequency in the document corpus.

Load the *wordcloud* package, as follows:

```
library(wordcloud)
```

For Word Cloud comprising terms with a frequency greater than 30, use the following command:

```
wordcloud(names(freq), freq, min.freq=30,colors=brewer.
pal(3,"Dark2"))
```

For a Word Cloud for the 50 words that occur most often, use the command given below:

```
wordcloud(names(freq), freq, max.words=50,colors=brewer.
pal(6,"Dark2"))
```

Text processing is about extracting useful information from text, which includes basic steps of pre-processing data, stemming the data, representing the corpus using the document term matrix and obtaining the associations between terms. R provides several libraries and functions to efficiently carry out these tasks. END

References

[1] *https://cran.r-project.org/web/packages/tm/tm.pdf*
[2] *https://cran.r-project.org/web/packages/wordcloud/wordcloud.pdf*
[3] *https://cran.r-project.org/web/packages/SnowballC/SnowballC.pdf*

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